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Evolution of extrafloral nectaries: adaptive process and selective regime changes from forest to savanna

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Abstract

Much effort has been devoted to understanding the function of extrafloral nectaries (EFNs) for ant–plant–herbivore interactions. However, the pattern of evolution of such structures throughout the history of plant lineages remains unexplored. In this study, we used empirical knowledge on plant defences mediated by ants as a theoretical framework to test specific hypotheses about the adaptive role of EFNs during plant evolution. Emphasis was given to different processes (neutral or adaptive) and factors (habitat change and trade-offs with new trichomes) that may have affected the evolution of ant–plant associations. We measured seven EFN quantitative traits in all 105 species included in a well-supported phylogeny of the tribe Bignonieae (Bignoniaceae) and collected field data on ant–EFN interactions in 32 species. We identified a positive association between ant visitation (a surrogate of ant guarding) and the abundance of EFNs in vegetative plant parts and rejected the hypothesis of phylogenetic conservatism of EFNs, with most traits presenting K-values < 1. Modelling the evolution of EFN traits using maximum likelihood approaches further suggested adaptive evolution, with static-optimum models showing a better fit than purely drift models. In addition, the abundance of EFNs was associated with habitat shifts (with a decrease in the abundance of EFNs from forest to savannas), and a potential trade-off was detected between the abundance of EFNs and estipitate glandular trichomes (i.e. trichomes with sticky secretion). These evolutionary associations suggest divergent selection between species as well as explains K-values < 1. Experimental studies with multiple lineages of forest and savanna taxa may improve our understanding of the role of nectaries in plants. Overall, our results suggest that the evolution of EFNs was likely associated with the adaptive process which probably played an important role in the diversification of this plant group.

Introduction

Extrafloral nectaries (EFNs) are morphological structures that secrete nectar and are not involved directly in pollination (Elias, 1983). These structures increase plant resistance against herbivores by attracting insects (mainly ants) that protect plant tissues (e.g. Oliveira & Freitas, 2004). Although the outcome of facultative ant–plant interactions is context specific (Bronstein, 1994), that is, dependent on biotic and abiotic factors in a population scale (Kersch & Fonseca, 2005), ants generally protect plants and decrease herbivory, contributing to a positive effect on fitness in different species (Rosumek *et al.*, 2009). Ant–plant–herbivore interactions represent a great model for micro-evolutionary (Heil & McKey, 2003) and macro-evolutionary studies focused on the potential evolutionary outputs of such interactions.

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Macroevolutionary studies on the quantitative variation of traits, including defensive plant traits, generally test two main hypotheses: (i) phylogenetic conservatism, and (ii) trade-offs among co-occurring traits within single taxa (Agrawal, 2007). Phylogenetic conservatism occurs when closely related species are more morphologically/ecologically similar than would be expected based on their phylogenetic relationship (related to phylogenetic signal *sensu* Blomberg *et al.*, 2003). In this context, significant phylogenetic signal does not imply inertia or constraint (Blomberg & Garland, 2002), and many different evolutionary processes could produce similar phylogenetic signals (Revell *et al.*, 2008). For example, low phylogenetic signal arises from conditions of strong stabilizing selection to a single optimum, as well as from conditions of strong regular divergent selection (Revell *et al.*, 2008). Therefore, if one's goal is to assess the evolutionary processes underlying the distribution of character states among taxa, an advisable approach is to try to fit alternative evolutionary models to the trait data, in association with information on the phylogenetic history of the group (e.g. Agrawal *et al.*, 2009a,b).

Two general models can be considered while trying to differentiate among evolutionary processes: (i) pure drift model based on Brownian motion (BM) and (ii) Ornstein–Uhlenbeck model (OU) incorporating both selection and drift (Butler & King, 2004). Expected evolutionary dynamics of potential adaptive traits (e.g. defensive traits as EFNs) should fit better OU models than purely neutral models. In studies in which the patterns of trait evolution do not follow a BM model, researchers seek to understand the factors (e.g. variation in the agents of selection during evolution) that might be responsible for the evolutionary deviations from the neutral model of evolution. Hypotheses on the potential agents of selection associated with these characters at microevolutionary scales might help refine hypothesis at macroevolutionary scales. The importance of this framework is even greater when the studied characters are related to a specific function and relevant processes within a plant's life cycle, such as pollination (e.g. Alcantara & Lohmann, 2011), seed dispersal (e.g. Lorts *et al.*, 2008) and plant defence against herbivores (e.g. Agrawal, 2007). However, detailed ecological data are generally lacking for the majority of species, complicating studies at macroevolutionary scales. For EFNs in particular, very few studies have explored their evolutionary history and/or examined their function in detail (but see Marazzi & Sanderson, 2010).

Habitat change can promote a modification in the assemblage of ants and herbivores that interact with plants. Such change in the defensive ant assemblage may, in turn, modify the cost/benefit of EFNs for plants, leading to deviations in the evolution of these secretory structures. For instance, if two sister species or clades are under different selective regimes, we expect their mean phenotypes to differ (Butler & King, 2004). In some ecological conditions such as the savanna environments, the occurrence of highly efficient herbivores (harmful to plants and indifferent to ants), the occupation of EFNs by nonprotective ants (e.g. Nogueira *et al.*, 2012) or the lack of ants (e.g. Moraes & Vasconcelos, 2009) may lead to a high cost for the production of EFNs, with the strength of selection against EFNs depending on the cost associated with these structures. Changes in the environment/habitat occupied by individual plants over time may lead to a replacement of the associated fauna (see Vasconcelos & Vilhena, 2006; Verhaag & Rosciszewski, 1994, in the case of ant assemblages replacement) and a decoupling of interactions, shifting the evolutionary direction of the nectaries that mediate these interactions.

On the other hand, ecological, genetic and developmental factors are thought to contribute to deviations from the Brownian mode of evolution of EFNs. Individual EFN characters are expected to interact more with characters located in the same plant parts (or organ) than with characters located in different plant parts (within and between different modules; Eble, 2005). Such interactions may be direct (i.e. genetic or ontogenetic, involving rearrangement of a character in the same region where they develop) or indirect (e.g. involving ecological factors). For example, the appearance of certain trichome types (e.g. adhesive glandular trichomes) could directly decrease herbivore damage affecting feeding behaviour of herbivores (Levin, 1973) and indirectly complicate ant-guard movement over plant surfaces. Both effects could modify the evolutionary trajectory of nectaries. The distance between the individual trichomes located in the shoot of four plant species (myrmecophytes) seems to select for ant visitors by worker body length (Davidson *et al.*, 1989). This effect could also occur in other groups, mainly in plants with big and dense trichomes or with glandular trichomes with sticky secretion, because both structures have the potential to modify drastically, if not prevent, ant movement over the plant surface.

In sum, two main factors could significantly impact, directly or indirectly, the evolution of EFNs and the ant-guarding system: (i) extrinsic factors such as biotic or abiotic changes associated with the geographic distribution of plants (e.g. habitat shifts or colonization from Amazon or Atlantic rainforest to xeric environments such as the Brazilian cerrados and caatingas) and (ii) intrinsic factors such as the emergence of new morphological structures in the same organ (or related ones) in which the nectaries are distributed (e.g. glandular and/or nonglandular trichomes).

The main objective of this study was to test adaptive hypotheses related to the evolution of extrafloral nectaries (EFNs), a special type of nectar-secreting glandular trichomes, in the tribe Bignonieae (Nogueira, 2011). Our overall adaptive hypothesis states that the evolution of quantitative EFNs traits represents a potential

defence against herbivores mediated by ant-guarding services, in which plants with more EFNs would be associated with higher levels of ant visitation. Under this hypothesis, we evaluated the phylogenetic signal of EFN traits to test the null hypothesis of trait phylogenetic conservatism (K-value > 1; Blomberg *et al.*, 2003) against alternative hypothesis (Revell *et al.*, 2008). We further predict that the adaptive model of macroevolution (OU model) should describe better the phenotypic divergence of EFNs during the phylogenetic history of the tribe Bignonieae than a purely drift model (Brownian motion).

We subsequently tested two additional and more specific hypotheses related to the adaptive evolution of EFNs: (i) the existence of an association between habitat shifts (from tropical rainforest to savanna-type environments, which represent new ecological scenarios, e.g. Nogueira *et al.*, 2012) and changes in EFN traits; and (ii) the existence of an association between the appearance of new morphological characters (different trichome types) and changes in EFN traits. In particular, an association between the invasion or occupation of savanna environments (forest > savanna) with the decrease or increase in EFN abundance would be interpreted as an evidence for divergent selection among contrasting habitats. In addition, an association between the emergence of new types of trichomes (glabrous > pilose) and a decrease in the abundance of nectaries would be interpreted as an evidence of trade-off between characters.

Finally, given that EFNs are of great importance for the establishment of variation in ant visits among plants, we used phylogenetic comparative methods to visualize the evolutionary pattern of EFN clusters on selected vegetative plant parts.

Materials and methods

Study system, phylogeny and data collection

The tribe Bignonieae (Bignoniaceae) mainly includes lianas that present an important ecological role in dry and wet neotropical forests (Gentry, 1991). The tribe includes 396 species (ca. 50% of all Bignoniaceae species) that are distributed through 21 monophyletic genera (Lohmann, 2006). A well-supported combined molecular phylogeny (i.e. 92% of the nodes with 50% or higher ML bootstrap support) including 104 species of Bignonieae based on plastid (ndhF) and nuclear (PepC) sequences is currently available for the group (Lohmann, 2006). This molecular phylogeny was combined with fossil data to estimate the time of divergence of all nodes in Bignonieae (Lohmann *et al.*, in press). The phylogeny used here was identical to the combined molecular phylogeny from Lohmann (2006) except that *Pyrostegia dichotoma* was treated as a synonym of *Pyrostegia venusta* and thus excluded from the tree, whereas *Callichlamys latifolia* and *Perianthomega vellosoi* were included to represent the only two genera of Bignonieae not included in the combined phylogeny of Lohmann (2006). The phylogenetic position of *Callichlamys* and *Perianthomega* followed relationships recovered in the ndhF phylogeny of Lohmann (2006), which included a broader sampling of taxa than the combined molecular (ndhF plus PepC) phylogeny of Lohmann (2006). Therefore, the phylogeny used here to study the evolutionary pattern of quantitative descriptors of extrafloral nectaries (EFNs) in Bignonieae included 105 taxa. This phylogeny is identical to that used to study the evolution of floral traits in Bignonieae (Alcantara and Lohmann, 2011). As we did not have *a priori* expectations for the patterns of evolution of EFN traits, branch lengths were untransformed.

The tribe Bignonieae has long been known for the high variation in the abundance and location of EFNs in the vegetative plant parts of the various taxa (Seibert, 1948; Gentry, 1991; Lohmann, 2003). These secretory structures are distributed on the nodes of the stems (interpetiolar fields *sensu* Gentry, 1991), prophylls of the axillar buds (*sensu* Lohmann, 2003), petioles, petiolules, adaxial and abaxial sides of leaflet blades (Seibert, 1948; Gentry, 1991; Lohmann, 2003). We sampled at least three specimens of each of the 105 species sampled in this study to describe the wide variation in EFN found in this group. To avoid spurious data from specimens, we selected recently collected plants whenever possible. Quantitative data from the extrafloral nectaries (nectar-secreting trichome in this group) and two other trichome types (sticky estipitate glandular and nonglandular) were obtained from the vegetative plant parts of dried herbarium specimens using a stereomicroscope and fresh materials collected in the field for some species. The exact sampling scheme adopted on different vegetative plant parts is described in detail in Appendix S1.

Relationship between ants and extrafloral nectaries

To study the relationship between the abundance of EFNs and ant visitation, we observed direct interactions between EFNs and ants on 32 species of Bignonieae from 32 km of pathways in four different Brazilian sites: (i) Reserva Ducke, Amazonas (rainforest); (ii) Parque Nacional do Viruá, Roraima (rainforest); (iii) Parque Estadual de Grão Mogol, Minas Gerais (savanna); and (iv) Parque Nacional da Chapada Diamantina, Bahia (savanna). We used trails from PPBIO grids (Reserva Ducke and Parque Nacional do Viruá – http://ppbio.inpa.gov.br/sitios) and common trails on savanna areas to sample ant–plant interactions. Ant–plant interactions were selected at least 10 m from each other to ensure that we were sampling interactions mediated by different ant colonies. Sample size was related to the abundance of each plant species and the abundance of ant–plant

interactions on each plant species. However, four individuals of each plant species were generally sampled (87% of species, and 53% with $N > 7$ individuals). Forest observations were conducted during the month of September (dry season), whereas savanna observations were carried out in November and February (rainy season). Although all the sampling should have ideally been conducted during the rainy season, this was not possible due to logistic limitations. However, differences in rainfall between seasons in Central Amazonian forests (where we concentrated 80% of the interactions sampled in forest habitats, the Reserva Ducke) are rather small. Therefore, we assumed that results from the sampling in the dry season in this site should not differ greatly from sampling in the rainy season. Plant species sampled with ants are presented in Appendix S2.

Because the active EFNs of Bignonieae are restricted to young branches and newly expanded leaves (e.g. Nogueira *et al.*, 2012), only plants with newly expanded leaves were sampled. To standardize the sampling effort across taxa, a single node from each plant was sampled in all cases. Each node includes an interpetiolar branch region (both sides), two prophylls and two pairs of leaves, with variable numbers of leaflets. The abundance of EFNs was quantified using two variables: (i) number of EFNs in the region with the greatest EFN cluster at each node and (ii) total number of EFNs per node (two opposite leaves, prophylls of the axillary buds, and interpetiolar region).

All ants observed visiting EFNs were counted on the newly expanded branches of each plant and then standardized by a single node for comparison between species. The main region for ant attraction in each species (EFN cluster) was documented. Ants were collected using special flat-tipped tweezers and fixed in 90% alcohol for subsequent identification. Ant collection was always conducted between 9 AM and 5 PM for all taxa. In addition, ant visitation was also monitored between 8 PM and 10 PM in order to identify possible replacements of ant species during this time period.

Linear regression testing by permutation was performed to examine the relationship between the abundance of EFNs and ant visitors across species. To fix the nonindependence of plant species, simple and partial Mantel tests were conducted on distance matrices taking into account the phylogenetic relationships between taxa. For this, a third matrix representing the phylogenetic distances between species (patristic distances) was calculated using the molecular phylogeny of Bignonieae (Lohmann, 2006) with the R package adephylo (Pavoine *et al.*, 2008). Although we observed 32 species interacting with ants, only 20 of these plant species were sampled in the Bignonieae phylogeny, and only these taxa were included in our analyses. Linear regressions and Mantel tests were performed with the R packages lmPerm (Wheeler, 2010) and Vegan (Oksanen *et al.*, 2011), respectively.

Phylogenetic signal and test of adaptive evolution of EFN traits

To test the null hypothesis of phylogenetic conservatism of EFNs (through a Brownian mode of evolution), we considered seven distinct characters to represent the quantitative variation in the abundance of EFNs between the 105 species sampled. For this, three dried herbarium specimens were sampled per taxon, and the average data were used in the analyses. Seven quantitative descriptors of EFNs were used: (i) EFNs in the interpetiolar region of branches (nodes), (ii) EFNs in the prophylls of the axillary buds, (iii) EFNs in the petioles and petiolules, (iv) EFNs on the intermediate zone of the adaxial side of leaflets, (v) EFNs on the base and apex of the adaxial side of leaflets, (vi) EFNs on the intermediate zone of the abaxial side of leaflets and (vii) EFNs on the base and apex of the abaxial side of leaflets. A multivariate analysis (PCA; principal components analysis) was used to reduce the dimensionality of the descriptors of EFNs and to concatenate the variation in abundance of EFNs into a smaller number of uncorrelated dimensions (three main axes). This approach was conducted due to expected dependence, structurally and ecologically (cost of EFNs on the whole plant × synergistic effect in attracting ant guards), among EFN plant variables (see a similar approach in Johnson *et al.*, 2009). More than that, this approach was used because some EFN variables (1, 2 and 3 above in this paragraph) presented a homoplastic pattern in the tribe Bignonieae (results not shown), preventing their use in subsequent hypotheses testing (e.g. tests of association detailed below).

According to the Brownian model of evolution, phenotypic variance is proportional to the time of divergence between species. We thus tested whether the descriptors of EFNs followed the Brownian mode of evolution through *K*-statistics (Blomberg *et al.*, 2003). This test was applied to the seven descriptors of EFNs and to the three principal components derived from the original variables. *K*-values not significantly different from zero indicate a lack of phylogenetic signal (random variation of the EFN traits). Whenever $K > 1$, EFNs were interpreted as being more similar in closely related species than expected by the Brownian mode of evolution (in accordance with the phylogenetic conservatism hypothesis). Whenever $K < 1$, EFNs were interpreted as being less similar in closely related species than expected by the Brownian mode of evolution.

To improve our understanding about the evolutionary processes behind the observed phenotypic divergence of nectaries and the *K*-values of the phylogenetic signal tests, we modelled the evolution of traits using

maximum likelihood approaches. We fitted and compared two different models of trait evolution, including Brownian motion and O-U processes with a single adaptive optimum (Butler & King, 2004). Models were chosen using the likelihood ratio test (LRT).

All analyses were conducted in the R packages Picante (Kembel *et al.*, 2010) and Geiger (Harmon *et al.*, 2009).

Tests of association between habitat, trichomes and EFNs

We tested two additional hypotheses related to the adaptive evolution of EFNs in the tribe Bignonieae: the potential effect of habitat change (forest > savanna) and the potential effect of the appearance of new trichome types (glabrous > pilose) on the number and distribution of EFNs on the plants. For that, both habitats and trichome types were coded as categorical variables.

To test the association between habitat changes and EFNs, three classes of habitats/environments were considered for each Bignonieae species: rainforest specialist, savanna specialist and habitat generalist (when individuals of a single species were found in both habitats). Specimens were classified as specialist in forest or savanna based on the habitat descriptions of Bignonieae species presented in Lohmann (2003). The savanna condition was used in a wide sense including general dry environments: cerrado vegetation, caatinga vegetation and transitional areas between both (e.g. carrascos). Definitions of cerrado and caatinga vegetations followed Eiten (1972).

To test for the association between the appearance of new trichome types and EFNs, three classes of trichomes were considered: (i) functionally glabrous (when nonglandular trichomes were absent or scarce and stipitate glandular trichomes were totally absent, not preventing ant movement or herbivore damage); (ii) with nonglandular trichomes (only providing a physical barrier to ant movement); and (iii) with adhesive stipitate glandular trichomes (providing both chemical and physical barriers to ants). Specimens were classified as having nonglandular trichomes whenever trichomes with sizes that were equal or greater than 0.3 mm were found on twigs and leaves, and/or whenever trichome density was equal or greater than 10 trichomes mm^{-2}. Nonglandular trichomes that were smaller or only present in densities that were smaller than those were considered functionally glabrous.

The effects of (1) habitat change and (2) trichome appearance were separately tested on the principal components (three axes), with each axis representing the variation of EFNs among taxa. Scores of species on each factor of the trait variation were used as dependent variables, whereas habitat and trichome appearance were used as categorical variables in a MANOVA (more than one dependent variable) and in an ANOVA (one dependent variable).

Both tests were controlled for phylogeny. The null distribution of this statistical test was obtained by simulating new sets of dependent variables on the phylogenetic tree under a Brownian mode of evolution. Whenever significant differences were found in the univariate analyses, *post hoc* tests were performed repeating the same procedure for each possible pair of habitat and trichome type. Standard and phylogenetic MANOVAS/ANOVAS were performed with the functions phy.manova and phy.anova of the R package Geiger (Harmon *et al.*, 2009).

Phylogenetic pattern of EFN clusters

To visualize the evolutionary pattern of EFN clusters on the phylogeny of Bignonieae, the quantitative data from EFNs were coded as discrete characters. Basing on our field observations, we considered five clustered EFNs as the minimum number of EFNs required to attract ants within a single location. We then coded the EFN traits of all 105 species of Bignonieae sampled as a binary character state in different plant parts: with 0 representing five or less EFNs; and 1 representing more than five EFNs in a single location (a cluster of EFNs). In general, the vegetative plant parts that most commonly presented EFN clusters were the base of the abaxial side of leaflets, the prophylls of the axillary buds and the interpetiolar region of stems (or nodes). Therefore, we reconstructed the ancestral character state of EFN traits in each of these vegetative plant parts.

Ancestral character state reconstructions were conducted using maximum likelihood assumptions and the software Mesquite 2.74 (Maddison & Maddison, 2007). In all analyses, EFN characters were equally weighted and considered unordered. Two evolutionary models were considered for the binary traits (EFNs): An Mk1 model with 1 parameter, in which any particular change is considered equally likely, and an Mk2 model with two parameters that represent forward and backward rates of change between character states. Models were chosen using the LRT.

Results

Abundance and clusters of EFNs and ant activity

Extrafloral nectaries were found in all species of Bignonieae studied in herbarium sheets ($N = 105$) and during fieldwork ($N = 20$). High variability in the abundance and position of the EFNs was observed among species (Fig. 1). The higher the abundance of active nectaries on vegetative plant parts (young nodes in expansion or newly expanded nodes), the greater the number of ants found on the nectaries (Fig. 2). The

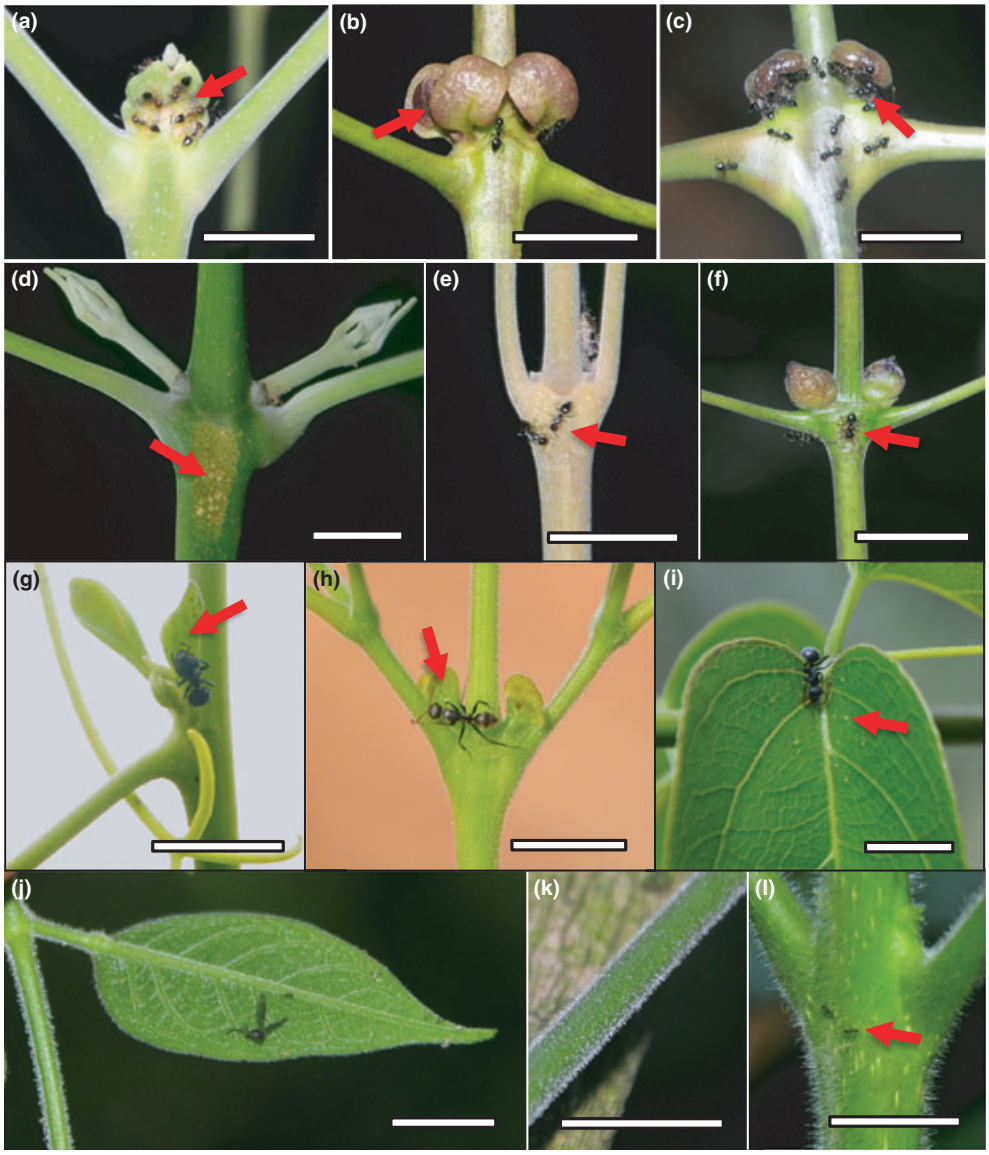


Fig. 1 Ant–plant interactions in species of Bignonieae from Neotropical rainforests and savannas. Red arrows indicate the location of nectar-secreting EFNs. (a–f) Amazonian taxa with *Crematogaster* ants visiting EFNs located on the prophylls of axillary buds (a–c) and interpetiolar portions of stems (d–f). (a) *Adenocalymma flaviflorum*, scale bar = 15 mm; (b) *Bignonia aequinoctialis*, scale bar = 12 mm; (c) *Anemopaegma robustum*, scale bar = 11 mm; (d) *Fridericia trailli* (secreting nectar), scale bar = 10 mm; (e) *Tynanthus panurensis*, scale bar = 10 mm; and (f) *Pachyptera aromatica*, scale bar = 16 mm. (g–i). Savanna species with ants visiting EFNs of the prophylls of the axillary buds (g–h) and the adaxial side of leaflets (i). (g) *Amphilophium elongatum* with *Cephalotes* ants, scale bar = 9 mm; (h) *Adenocalymma pedunculatum* with *Camponotus* ants, scale bar = 13 mm; and *Cuspidaria convoluta*, scale bar = 11 mm. (j–m) Trichomes on the plant surface: (j) Diptera stuck in glandular trichomes (with sticky secretion) on the leaflets of *Adenocalymma adenophorum*, scale = 8 mm; (k) glandular trichomes on the petioles of plants without ants, scale bar = 8 mm; and (l) nonglandular trichomes and scarce ants of the genus *Asteca* in *Fridericia* sp., scale bar = 21 mm.

positive relationship between ant abundance and the abundance of EFNs was maintained when the analysis was controlled for phylogeny (Table 1), considering only the abundance of clustered nectaries in particular positions per node. These results suggest that the differences observed in the number of ant visitors between species are mainly due to the presence of EFN clusters, with minor importance to the variation in EFNs scattered over the plants. In general, EFN clusters (> 5 EFNs) were observed in the same position for each

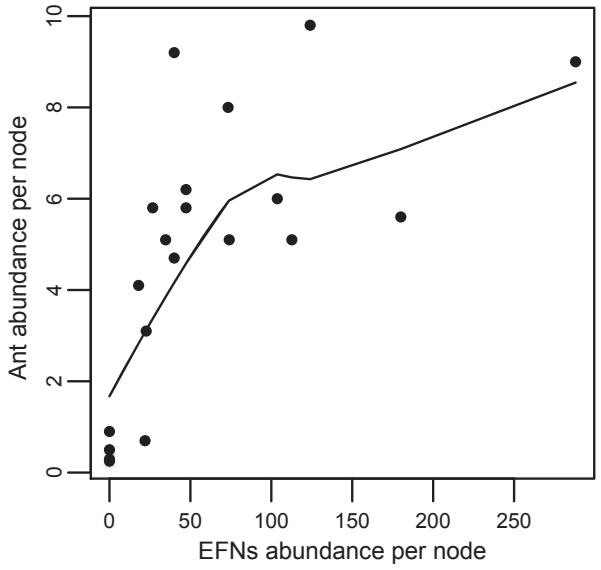


Fig. 2 Relationship between ant abundance and the number of EFNs in the region with the greatest EFN cluster of rainforest ($N = 17$) and savanna ($N = 3$) species of Bignonieae. Linear statistical results are presented in Table 1 (line 4). The trend line describing a visual relationship between the two variables was based on the lowest smoother using a locally weighted polynomial regression. The three species with fewer ants per node were savanna species. A complete list of the species sampled during fieldwork is available in the supplementary material (Appendix S2).

species, mostly in the prophylls of the axillary buds (22.8% of the species sampled), interpetiolar regions (36.2% of the species sampled) and at the base of the abaxial side of leaflets (37.1% of the species sampled). Less frequently, taxa presented two EFN cluster locations, in which the clusters appeared on the prophylls of the axillary buds, and at the abaxial side of leaflets (10.5% of species; e.g. *Anemopaegma robustum* and *Bignonia* spp.). Clusters of EFNs were also occasionally found at the base of the petioles (e.g. *Callichlamys latifolia*), petiolules (e.g. *Pachyptera kerere*) or at the apex of the abaxial side of leaflets (e.g. *Bignonia corymbosa*). Extrafloral nectaries did not present any clustered pattern in 23 of the 105 species sampled (21.9% of the species sampled). In general, EFNs were more abundant on the stems (interpetiolar region), prophylls of the axillary buds and base of abaxial side of leaflets (Table 2, second column).

A single species of ants generally monopolized the available EFN resources in a given plant, regardless of whether the plant was found in a forest or a savanna. In forests, the same ant species that used EFN resources during the day were also observed visiting the EFNs during the night. The average abundance of ants visiting plants in forests was 6.25 ants per node (SE = ± 0.93), with more than 50% of all interactions being observed in forests monopolized by ants of the genus *Crematogaster* ($N = 25$ plant species). In contrast, a turnover of ant species was observed between day and night periods in the savannas, and generally, different species of *Camponotus* were observed subsequently between these periods. In addition, a lower abundance of ants was observed interacting with EFNs in savannas, with an average of 0.53 ants per node (SE = ± 0.12). Differently from forest species, the most frequent and abundant group of ants on savanna plants was *Camponotus* ($N = 7$ plant species). Further details on the abundance of ants are presented in Appendix S2.

Table 1 Relationship between ant abundance, density of EFNs and phylogenetic distance (phy-distance) among 20 species of Bignonieae. Phylogenetic distance estimation (patristic distance) was used to control the analysis for phylogenetic relationships. Variables in all Mantel tests were converted to Euclidean distances. EFN1 represents the abundance of clustered nectaries in particular positions per node; EFN2 represents the total abundance of nectaries per node. Statistical tests with $P < 0.05$ level of significance are presented in bold.

	Relationship among variables	Statistical test	*r*	*P*
1	Ant abundance to phy-distance	Simple Mantel	0.02	0.34
2	EFN1 to phy-distance	Simple Mantel	−0.08	0.67
3	EFN2 to phy-distance	Simple Mantel	0.01	0.48
4	Ant abundance to EFN1	Simple Regression	**0.64**	**0.002**
5	Ant abundance to EFN2	Simple Regression	**0.52**	**0.018**
6	Ant abundance to EFN1 controlling for phy-distance	Partial Mantel	**0.29**	**0.01**
7	Ant abundance to EFN2 controlling for phy-distance	Partial Mantel	0.12	0.18

Test of phylogenetic conservatism and evolutionary processes

The first three axes of the PCA conducted with seven descriptors of the variation of EFNs explained more than 60% of the variation in the original data (Table 2). The first axis mainly expressed the variation in the abundance of EFNs in the prophylls of the axillary buds ($r = -0.739$), and of EFNs scattered on the adaxial ($r = -0.663$) and abaxial sides of leaflets ($r = -0.699$). The second axis, on the other hand, expressed the variation in the abundance of EFNs in the interpetiolar portions ($r = -0.792$), whereas the third axis expressed the variation at the base of the adaxial side of leaflets ($r = -0.620$).

K-statistics were significantly different from zero for four EFN characters and for the first two axes of the PCA (Table 3), indicating that the evolution of these characters was not random during the history of the

Table 2 Descriptive data of EFN variables in each plant part among 105 Bignonieae species and the PC1, PC2 and PC3 loadings from principal component analyses. Loadings higher than 0.4 are highlighted in bold. The proportion of variation explained by the original data of each principal component is presented in parenthesis. EFNs clustered on leaflet blades represent the sum of EFNs on the apex and base of leaflets, where these clusters were common on the abaxial surface of leaflets.

				Principal components		
Characters	Position of EFNs on the plants	% of species without EFNs	Average abundance of EFNs (max–min)	PC1 loadings (26%)	PC2 loadings (24.1%)	PC3 loadings (16.2%)
1	Stem (node)	53.8	27.7 (0–288)	0.263	**−0.792**	0.034
2	Prophylls of the axillary buds	60.4	5.4 (0–64)	**−0.739**	**0.488**	−0.091
3	Petiole and petiolule	72.6	3.3 (0–120)	0.054	**−0.566**	**0.465**
4	Adaxial side of leaflets (intermediate zone)	37.7	0.5 (0–5)	**−0.663**	−0.378	−0.315
5	Adaxial side of leaflets (base and apex)	28.3	3.9 (0–115)	−0.315	**−0.568**	**−0.620**
6	Abaxial side of leaflets (intermediate zone)	9.4	1.1 (0–6)	**−0.699**	−0.155	**0.425**
7	Abaxial side of leaflets (base and apex)	2.8	10.4 (0–103)	**−0.426**	−0.146	**0.488**

group. Three of these four EFN characters were associated with plant parts having clusters of the EFNs (probably the most important regions for the attraction of ants on these plant species) and/or showing the greatest range of variation in the abundance of EFNs (Table 2). All values of K significantly different from zero were lower than 1 ('low values of K'), indicating that the divergence in the abundance of EFNs among closely related taxa was different from the expected pattern of neutral evolution, and probably greater than expected by the shared history among these species (Table 3). In other words, those characters did not present phylogenetic conservatism.

The evolutionary pattern of EFNs was more appropriately described by an adaptive model of evolution than by Brownian model for almost all quantitative traits considered in this study (Table 3). Five of six single EFN traits, and two of three PCA axes (second and third axes), were best fitted by a static-optimum model (O-U process), which often is interpreted as indicating stabilizing selection.

Association between habitat shifts, trichomes and EFNs

To explore additional processes and factors that might be associated with the low values of K (i.e. higher divergence than expected by a neutral mode of evolution), we tested whether habitat shifts (e.g. transitions from forest to savanna) presented a negative impact on the evolution of EFNs (e.g. decrease in abundance of nectaries), causing deviation from the Brownian mode of evolution. The first PCA axis, which considered mainly the variation of EFN clusters in the prophylls of the axillary buds, was associated with change in plant habitat (Table 4). More specifically, savanna species presented a decrease in the number of nectaries when compared to the forest taxa (Fig. 3). Whereas habitat generalists and forest specialists did not differ in the first axis of the PCA, savanna species were much different from those groups. However, this pattern was not as clear in the second axis of the PCA, the axis that represented the EFN clusters in the interpetiolar regions (Fig. 3). In this axis, forest specialists and savanna specialists were not statistically different from each other, although the savanna taxa presented fewer EFNs in the interpetiolar regions on average when compared with forest specialists and habitat generalists. In particular, habitat generalists presented a higher number of EFNs when compared with the other two groups.

As far as the evolution of glandular and nonglandular trichomes is concerned, only the emergence of stipitate glandular trichomes presented a clear effect on the evolution of the abundance of EFNs (Table 4), leading to a drastic reduction in EFNs on the plants (Fig. 3). The effect of the evolution of this kind of glandular trichomes on the reduction in EFNs was clear in the tribe Bignonieae despite the small sample size ($N = 9$ species considered, and probably nine independent evolutions of this kind of trichome in the tribe Bignonieae).

Ancestral character state reconstructions

The macroevolutionary pattern of nectaries estimated by ML ancestral character state reconstructions indicated that the most recent common ancestor of the tribe did not present clustered EFNs in any plant parts (Fig. 4 and Appendix S3). On the other hand, the most recent common ancestor of the tribe likely presented low amounts of EFNs scattered over the leaf blade with estimated probability of presence (p.p.) equal 0.75 (without statistical support, Appendix S3). In addition, EFNs were probably lacking from the interpetiolar region (p.p. = 0.03) and prophylls of the axillary buds

Table 3 Phylogenetic signal and modelling of abundance of EFNs distributed on different plant parts of 105 species of Bignonieae (Bignoniaceae). *K*-statistics are given for each trait separately and for each principal component (second column). Traits significantly associated with phylogenetic history are presented in bold ($P < 0.05$; third column). Modelling of the mode of evolution was described in the last three columns and the best-fitted model is presented in bold. EFN trait 3 was probably absent in the most recent ancestor of the tribe Bignonieae (results not shown), so that they were not modelled independently by Brownian motion and Ornstein–Uhlenbeck (O-U) models (NA, not applicable). See appendix S1 for detailed description of traits and abbreviations.

Evolutionary model	*K*-statistic	*P*-value	Model parameter MLE	LL	AIC
1 – Stem (node)*					
Phylogenetic signal	0.64	**0.012**			
Brownian motion				−706.7	1417.5
O-U			$\alpha = 0.049$	−533.8	**1073.9**
2 – Prophylls of the axilary buds*					
Phylogenetic signal	0.67	**0.002**			
Brownian motion				−380.0	764.1
O-U			$\alpha = 0.025$	−377.9	**762.1**
3 – Petiole and petiolule*					
Phylogenetic signal	0.27	0.915			
Brownian motion				NA	NA
O-U			$\alpha = NA$	NA	NA
4 – Adaxial side of leaflets (int.)					
Phylogenetic signal	0.68	**0.001**			
Brownian motion				−112.1	**228.4**
O-U			$\alpha = 0.006$	−112.0	230.3
5 – Adaxial side of leaflets (b and a)					
Phylogenetic signal	0.35	0.690			
Brownian motion				−417.3	840.9
O-U			$\alpha = 0.269$	−393.6	**793.5**
6 – Abaxial side of leaflets (int.)					
Phylogenetic signal	0.37	0.747			
Brownian motion				−186.2	376.6
O-U			$\alpha = 7.426$	−164.6	**335.5**
7 – Abaxial side of leaflets (b and a)					
Phylogenetic signal	0.68	**0.009**			
Brownian motion				−423.8	851.8
O-U			$\alpha = 0.048$	−417.3	**840.9**
PCA1 (26% of EFN variation)					
Phylogenetic signal	0.70	**0.001**			
Brownian motion				−162.5	**329.2**
O-U			$\alpha = 0.022$	−160.9	328.0
PCA2 (24% of EFN variation)					
Phylogenetic signal	0.57	**0.002**			
Brownian motion				−167.9	339.9
O-U			$\alpha = 0.049$	−159.5	**325.2**
PCA3 (16% of EFN variation)					
Phylogenetic signal	0.46	0.242			
Brownian motion				−160.6	325.3
O-U			$\alpha = 0.129$	−146.3	**298.8**

*Homoplastic pattern (the appearance of EFNs on this plant part evolved more than once in the history of the tribe Bignonieae; results not shown).

(p.p. = 0.02). However, the first evolution of clusters of EFNs in the tribe Bignonieae (> 5 EFNs in the same position of the plant) seems to have happened early in the evolution of the group. The first plant part that probably presented a large amount of clustered EFNs was the prophyll of the axillary buds. Detailed results of the ancestral character state reconstructions for all characters are available in Appendix S3.

Discussion

This study used the hypothesis of defence mediated by ants and ant guarding (Heil & McKey, 2003) as our basic theoretical framework and tested a series of specific hypotheses associated with the potential adaptive role of extrafloral nectaries during the evolution of the tribe Bignonieae (Bignoniaceae). Emphasis was given

Table 4 Impact of habitat change (extrinsic factors) and impact of the evolution of trichomes (intrinsic factors) on the evolutionary history of EFNs based on phylogenetic MANOVA/ANOVA tests of association. The variation in EFNs was represented by scores on principal component analyses conducted on variables representing the abundance of EFNs on different plant parts (Table 2). Habitats of Bignonieae species were classified as tropical rainforest, dry areas or both (generalist behaviour); trichomes were classified as functionally absent, with nonglandular trichomes (physical barriers only) or adhesive glandular trichomes (physical and chemical barriers). Effects with $P < 0.05$ level of significance are presented in bold. Marginally significant effects are also noted (+).

EFN variables	*F* statistic	Traditional *P* value	*P*-value corrected by phylogeny (*P**)
Habitat change effect			
MANOVA test ($F_{(2,97)} = 3.18$; $P = 0.005$; $P^* = 0.001$)			
PCA1 (EFNs in prophylls and leaflet blades)	4.92	0.066	**0.058 (+)**
PCA2 (EFNs in stems)	7.90	0.008	**0.007**
PCA3	1.87	0.193	0.243
Appearance of trichome effect			
MANOVA test ($F_{(2,97)} = 1.09$; $P = 0.365$; $P^* = 0.287$)			
PCA1 (EFNs in prophylls and leaflet blades)	2.50	0.087	**0.066 (+)**
PCA2 (EFNs in stems)	0.02	0.982	0.980
PCA3	0.98	0.379	0.324

to different processes (e.g. neutral or adaptive) and factors (i.e. habitat change and appearance of new trichome types) that may have affected the evolution of ant–plant associations. Historically, detailed experimental studies with natural populations indicated that ants can act as agents of selection on individual EFN traits (e.g. Rudgers, 2004). Indeed, plants with higher rates of ant visitation are less visited by herbivores and suffer less leaf damage, leading to a greater production of flowers and seeds (Rosumek *et al.*, 2009; Trager *et al.*, 2010). In other words, the frequency of nectaries is positively correlated with seed production (Rosumek *et al.*, 2009; Trager *et al.*, 2010). Such evidence supports the hypothesis that facultative and generalist ant–plant interactions may have played important roles for the evolution of nectaries. In this context, temporal and spatial variation in the interactions between biotic agents (ants and herbivores) are thought to have been involved in the evolution of EFNs between species over long periods of time. However, such microevolutionary patterns had not been previously linked to expected patterns of the macroevolution of EFNs in plants. In this study, we encountered an association of the abundance and clustered patterns of EFNs in vegetative plant parts with ant visit activity (a surrogate of ant guarding). Moreover, we documented the macroevolutionary patterns of EFN abundance, which was non-neutral but adaptive, associated with habitat shifts and influenced by trade-offs with other defensive traits (i.e. sticky glandular trichomes).

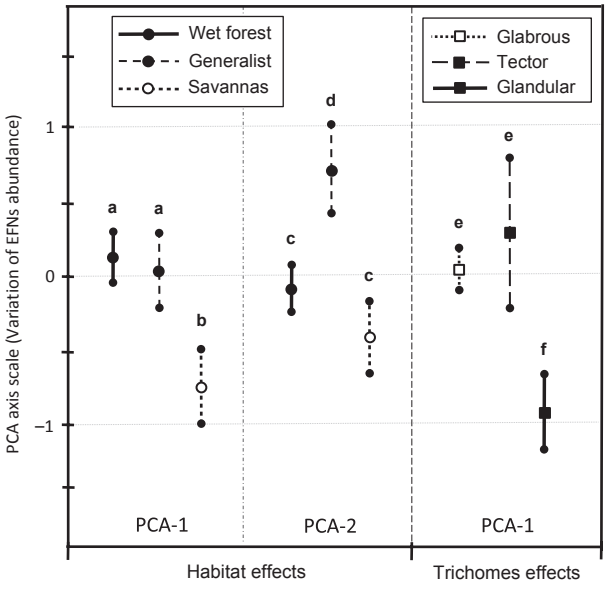


Fig. 3 *Post hoc* tests performed for each possible habitat pair (forest specialists, habitat generalists and savanna specialists) and trichome types (glabrous, tector and glandular) after finding differences in the univariate analyses (PCA-1 and PCA-2 for habitat changes; and only PCA-1 for the evolution of trichomes). Three independent analyses were performed for each PCA axis. Different letters are used to indicate significant differences between treatments ($P < 0.05$) in each analysis.

Relationship between ant activity and the abundance of EFNs

Nectaries of representatives of the tribe Bignonieae differ from those from other plant groups (e.g. *Inga*, *Senna* – Marazzi & Sanderson, 2010; *Chamaecrista* – Rutter & Rausher, 2004) by their small size. More specifically, these nectaries are patelliform or cupular glandular trichomes (Elias, 1983; Nogueira *et al.*, unpublished) and are distributed in various arrangements and patterns of abundance over various plant parts (Lohmann, 2003; A. Nogueira, pers. obs.). In this study, the pattern of

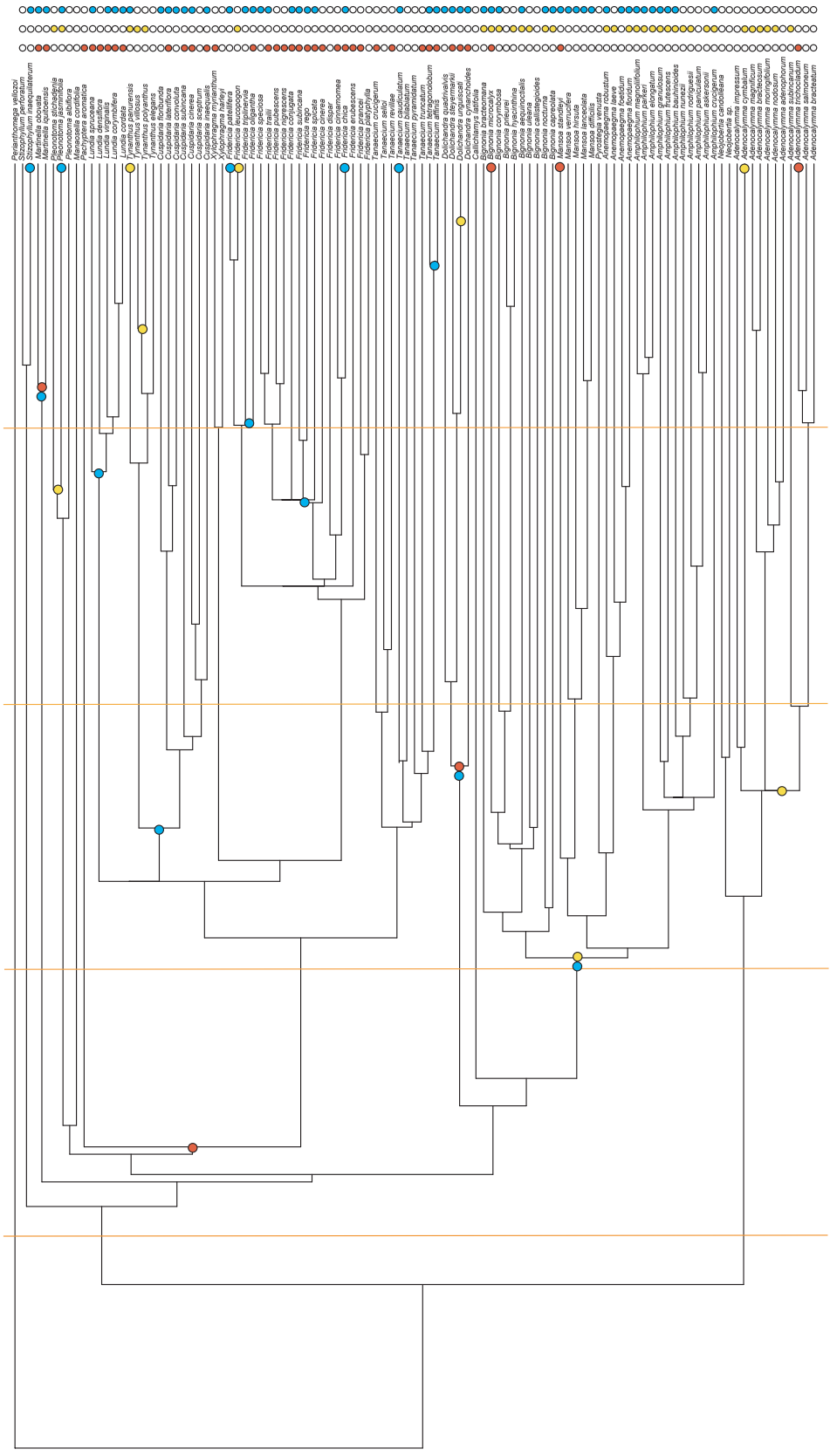


Fig. 4 Evolution of clusters of extrafloral nectaries (>5 EFNs). The evolutionary appearance of EFN clusters (proportional likelihood of appearance > 80%) is presented in red (interpetiolar regions), yellow (prophylls of the axillary buds) and blue (leaflet base) on the phylogeny of Bignonieae (Bignoniaceae) with branch lengths proportional to time (my) presented in Alcantara & Lohmann (2011). Losses of these characters and the detailed ancestral character reconstructions are presented in Appendix S3. Coloured circles next to species names represent the EFN character state in each plant part.

quantitative variation of EFNs among species was positively correlated with the number of ant visitors (i.e. ant visits reached their maximum in plants with the higher amounts of EFNs). In addition, species with higher ant abundance also seemed to be less damaged by herbivory (A. Nogueira, personal observaion). Nonetheless, a careful examination of such trend would require periodic plant surveys for a high number of species.

Extrafloral nectaries presented their highest concentration (cluster EFNs) in the prophylls of axillary buds, interpetiolar regions of branches and base of the abaxial side of leaflets. These vegetative plant parts were critical for the maintenance of ant attraction in plants. This is the first study to record such correlation among species, representing an important step towards a better understanding of the patterns of ant attraction to different types of EFNs (Díaz-Castelazo *et al.*, 2005). In two species of *Anemopaegma*, the quantitative variation of EFNs was associated with ant visitation (Nogueira *et al.*, 2012). In particular, the abundance of ants was greatest when the average number of nectaries on the leaves (and/or the number of leaves with active nectaries) was also higher. Field studies have often investigated the relationship between the number of visitors to EFNs and the identity of the species of ants that visit those EFNs (e.g. Rico-Gray & Oliveira, 2007). However, studies that provide information about the abundance of ant visitors (or the most frequent ant visitors) are still scarce. The last quantitative reviews on the subject did not present a detailed account on the relationship between EFNs and ants and did not provide information on the average number of ants per plant (or plant parts) in the taxa of interest (Chamberlain & Holland, 2009; Rosumek *et al.*, 2009; Trager *et al.*, 2010).

It is widely accepted that herbivore protection is directly proportional to the degree of ant-related investment by plants (Stanton & Palmer, 2011). Despite the importance of ant abundance to plant protection (Bronstein, 1998), this variable had been often neglected (e.g. review in Trager *et al.*, 2010). Although the diversity of ants is extremely important for ant–plant interactions, the actual number of ant visitors is also critical to reduce herbivory (e.g. Rudgers & Strauss, 2004). In other words, to measure (and justify) the success of an interaction we need to know 'who are the ant visitors?' and also 'how many ants are there?' (Bronstein, 1998). In the tribe Bignonieae, forest plant species were shown to attract a higher number of ants than the savanna species. Whereas the savanna species mostly attracted *Camponotus* ants, forest species mainly attracted *Crematogaster* ants (see Appendix S2), the most dominant ant-type in terms of resource use in forest environments (Baccaro *et al.*, 2010). The differential ant identity and abundance found in the different environments studied here was not unexpected because a high turnover of ant species has been described in nearby savannas and forests (e.g. Verhaag & Rosciszewski, 1994; Vasconcelos & Vilhena, 2006).

EFN phylogenetic conservatism and evolutionary processes

Extrafloral nectaries are known to be evolutionarily labile (Rudgers & Gardener, 2004; Heil, 2008, 2011). However, very few studies have actually addressed the evolution of EFNs using a robust phylogenetic framework. In this study, *K*-values for EFN descriptors (and principal components representing them) were shown to be lower than 1. These results indicated a greater phenotypic divergence between sister species than expected according to a neutral mode of evolution (i.e. Brownian mode of evolution), corroborating the high evolutionary lability and the absence of phylogenetic conservatism of nectaries. Likewise, a $K < 1$ also indicated the evolution of similar morphologies (e.g. clusters of EFNs) in distantly related taxa (Blomberg *et al.*, 2003; Revell *et al.*, 2008). This pattern also suggests that common ecological and evolutionary forces may have driven the morphological diversification of distantly related lineages within Bignonieae. Evolutionary lability (or low phylogenetic signal) implies rapid evolutionary change but does not specify a biological process (e.g. genetic drift, natural selection; Revell *et al.*, 2008).

Apart from assessing the phylogenetic signal (and estimating *K*-values), we also used a model selection approach to differentiate the evolutionary processes that were most likely associated with the evolution of EFNs (e.g. Agrawal *et al.*, 2009b). The adaptive model of macroevolution (O-U model with a single optimum) described better the phenotypic divergence of EFNs among Bignonieae species than a purely drift model (Brownian motion). The better fit of this model helps to explain the *K*-values < 1 obtained. In particular, those results indicate that stabilizing selection was important in the evolution of EFNs in this plant group. This interpretation is in agreement with the model of cost and benefit related to ant–plant interactions proposed by Fonseca (1993), and revisited by Bronstein (1998).

Habitat shifts, trade-offs and the evolutionary lability of EFNs

In this study, multiple evolutions of clusters of EFNs were documented in different plant parts of representatives of the tribe Bignonieae, suggesting a recurrent selective force on these structures in this clade. Such lability may have been caused by the beneficial role of the interaction with ants, leading to an increase in plant resistance against herbivores. In general, studies considering the conditionality of ant–plant interactions have shown that these interactions are mostly

beneficial, leading to a significant reduction in herbivory and a significant increase in characteristics that are associated with plant fitness (Chamberlain & Holland, 2009). However, most of the studies on ant–plant interactions did not include phylogenetically controlled experiments, limiting the value of such results. Here, we considered a series of testable hypotheses about factors that could potentially modify the cost–benefit of EFNs in plants during evolutionary time. In particular, any reduction in the benefits (e.g. absence or reduction in protective ants) and increase in costs (e.g. undesirable water loss) could induce a counter-selection of EFNs. In turn, any reduction in costs and/or increase in benefits of EFNs could generate selection towards an increase in the abundance of EFNs on the presence of genetic variation within populations, thus leading to the establishing of these characters in evolutionary lineages.

Tests of the association between habitat (changes from forest to savanna) and trichomes (changes from glabrous to glandular pubescent) on the evolution of the abundance of EFNs indicated that both factors may alter the evolutionary pattern of EFNs expected by the null model (neutral evolution). This association could be an evidence to divergent selection between sister clades or species occupying different environments (or with different trichome types). K-values < 1 have also been associated with strong divergent selection between species (Revell *et al.*, 2008) and might also explain the lack of phylogenetic conservatism in EFN traits encountered here.

It is reasonable to assume that the benefits of EFNs in drier environments would be considerably lower in species of Bignonieae, given that changes of environment/habitat have an impact on the relationships between plants and ant visitors, and between plants and herbivores. For example, we observed quantitative (i.e. a decrease in the number of ants per plant node) and qualitative (i.e. an increase in interactions with *Camponotus* ants) changes in ant visitation to EFNs after transitions from forests to savannas. Indeed, the appearance of very aggressive herbivores in savannas was documented in savanna species of Bignonieae, whose immediate ancestors occupied forest environments (Nogueira *et al.*, 2012). Moreover, changes associated with the water consumption by EFNs would likely make these structures more costly in savannas than in the original rainforest environments. More specifically, a reduction in the number of EFNs would be expected to balance the cost–benefit of EFNs in lineages of plants that invaded drier environments.

The first evolutionary event, and most of the other evolutionary transitions of EFN clusters within the tribe Bignonieae, likely took place in forest environments, perhaps because there has not been sufficient time since the first occupations of savannas for these traits to evolve within this environment. More specifically, the majority of the evolutions of clusters of EFNs (regardless of the position in the plant) seem to have occurred in lineages that occupied forests (biogeographical hypothesis of habitat occupation/transitions in Bignonieae from Lohmann *et al.*, submitted). In other words, the events of habitat transition were probably much younger in this plant group than the evolution of clusters of EFNs (as well as the appearance of glandular trichomes). It is possible that the cost–benefit balance between ants and EFNs in the forest may have been positive to plants in multiple lineages independently (i.e. those interactions would present a low cost and be highly protective against herbivores). However, experimental tests on the biological function of EFNs in forest species of the tribe Bignonieae would be necessary to corroborate this hypothesis. Functional data for EFNs are extremely rare. For the Bignoniaceae, these data are only available for three North American species belonging to *Campsis* and *Catalpa* (Elias & Gelband, 1976; Stephenson, 1982; Ness, 2003) and two savanna species belonging to the genus *Anemopaegma* (Nogueira *et al.*, 2012). Further functional studies are greatly needed for EFNs in savanna and forest environments.

Similarly to what happened with habitat shifts, the evolution of stipitate glandular trichomes with sticky secretion seems to have also contributed to a deviation from a neutral pattern of evolution of EFNs. It is possible that the evolution of these trichomes may have altered the pattern of EFN use by ants and the degree of resistance of vegetative plant parts. Those results explain the reduction in EFNs in plants presenting these structures, a first evidence of trade-off between different defensive traits. However, there is no empirical evidence for inter-specific trade-off among these defensive traits, a relationship of competing demands between EFNs and other trichomes types in Bignoniaceae. Thus, the correlative evidence presented here is based on the assumption of competing demands and awaits selection experiments for confirmation.

No ant visitors were observed in two species of Bignonieae with adhesive glandular trichomes, *Adenocalymma adenophorum* (forest) and *Cuspidaria sceptrum* (savanna). A similar pattern was seen in the genus *Chamaecrista* (Leguminosae), in which the loss of EFNs occurred concurrently with the evolution of adhesive glandular trichomes (Conceição *et al.*, 2009). The patterns observed in *Chamaecrista* and Bignonieae suggest a disruption of the ant–plant relationship, with the glandular trichomes making the movement of ants more difficult over the plants. In addition, the emergence of a new coverage of adhesive glandular trichomes could directly increase plant resistance and decrease herbivore damage (Levin, 1973). It is also possible that plants with this type of trichomes might have been associated with the arrival of new bodyguards that could, in turn, be more effective for plants than ants. Spider–plant interactions have been documented in *Trichogoniopsis adenantha*, in which spiders from the genus *Peucetia*

effectively defended those glandular plants against herbivores (Romero *et al.*, 2008). In the lack of live preys, the spiders fed on insect carcasses that were attached to adhesive glandular trichomes. This pattern was observed in another glandular plant species, *Rhynchanthera dichotoma* (e.g. Morais-Filho & Romero, 2010).

Concluding remarks

In the tribe Bignonieae, the decrease in EFN abundance did not lead to a complete disappearance of such structures in the group. In fact, EFNs were observed in all 105 species sampled. However, a drastic decrease in the abundance of such structures was observed in some lineages (i.e. some species of the genus *Anemopaegma*, *Cuspidaria* and *Fridericia*). This pattern suggests that the presence of a few scattered (not clustered) nectaries over the plants might not represent a significant cost for the plants (and probably no benefit). On the contrary, the maintenance of few scattered nectaries could predispose lineages with few nectaries to retrieve the number of 'lost' nectaries in the presence of additive genetic variability in this feature and selective forces in favour of this mechanism of anti-herbivory defence.

Additional studies on the evolutionary ecology of EFNs in plant groups with discrete and/or quantitative variation in EFNs would be extremely important to determine whether the patterns here encountered can be generalized to other plant lineages. Even though a series of transitions from forests to savannas are known to have happened in the tribe Bignonieae (Lohmann *et al.*, submitted) and other plant groups (Simon *et al.*, 2009), the opposite scenario was documented in *Rhipsalis* (Cactaceae), in which habitat transitions seem to have occurred from drier areas to humid forests (Calvente *et al.*, 2011). Lineages that have historically shifted their habitats are particularly useful for a better understanding of the patterns of evolution of plant–animal interactions in macroevolutionary scales.

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Supporting information

Additional Supporting Information may be found in the online version of this article:
Appendix S1 Sampling scheme of EFNs and trichomes in different vegetative plant parts.
Appendix S2 Ant abundance sampled during fielwork, including species from savannas (State Park of Grão Mogol – MG), and wet forests (National Park of Viruá – RR, and Ducke Reserve – AM).
Appendix S3 ML ancestral character state reconstructions of EFN clusters in the tribe Bignonieae (Bignoniaceae).

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